Mapping and Quantifying Engagement Levels of Governments in IPCC Approval Meetings by Topic from Text

Abstract

We introduce a text-as-data approach from natural language processing that allows mapping and quantifying *topic-level* variation in engagement levels of member governments in intergovernmental meetings. Building on research that characterizes the Intergovernmental Panel on Climate Change (IPCC) approval process as a key site for legitimating knowledge input into negotiations under the United Nations Framework Convention on Climate Change (UNFCCC), we illustrate this method for two key topics from the IPCC's Sixth Assessment Report (AR6) that became identified policy approaches for national mitigation and adaptation plans in the negotiated Global Stocktake outcome of COP28: "carbon management" and "nature-based solutions." Our analysis provides robust evidence that governments' engagement levels vary by topic, which allows us to nuance existing findings that IPCC meetings are dominated by a few large country delegations across all agenda items. Our proposed method can be applied to other intergovernmental meetings and complements qualitative research methods.

Keywords: IPCC; Sixth Assessment Report; UNFCCC; intergovernmental negotiations; text-as-data; word embeddings; carbon management; carbon capture and storage; carbon dioxide removal; nature-based solutions; ecosystem-based adaptation

Introduction

The release of a new report by the Intergovernmental Panel on Climate Change (IPCC) is a pivotal event for the international climate science and policy-making community. This is more true than ever as the Paris Agreement recognizes the IPCC as a source of the "best available science" to inform the 5-yearly Global Stocktake of the implementation of the agreement [1, 2]. While such heightened attention comes with political contestation over the alignment of IPCC and UNFCCC processes—as we see at present in discussions over the timeline of the Seventh Assessment Report (AR7)—it also highlights the importance of research that conceptualizes the IPCC's approval practice as a key site for crafting and legitimating the role of IPCC knowledge input into the UNFCCC negotiations [3–7]. We contribute to this research program by showing that governments do not engage uniformly in IPCC approval meetings, but that engagement depends on the topics under approval. These findings nuance existing evidence that a small group of influential states dominates discussions in the IPCC across all agenda items [5, 8–10]. We show instead that governments engage selectively on *some* topics, but not on others. This matters for two reasons: first, for documenting which governments engage on particular topics in IPCC proceedings; and second, for deepening our understanding of how such varied, topic-specific engagement levels in the IPCC are related to government positions on the *same topic* in the UNFCCC negotiations.

Key findings of IPCC reports are presented in the Summary for Policymakers (SPM) after member governments' line-by-line approval. Accounts of these closed meetings provide insight into national government attempts to negotiate, rewrite, or remove sentences, figures and whole sections of the summary [11]. The majority of these observer accounts have been provided by IPCC authors whose role is to ensure that the SPM is consistent with the underlying science assessed in the main report [12–14].

Social science analysis of these events has traditionally been based on ethnographic study [7, 8, 15–17], including by researchers embedded within delegations [3, 18, 19]. This research has provided the scholarly foundation for unpacking the relationship between science and politics in the approval process of the IPCC and, in some cases, indicated the impact of named countries on the final version of the SPM [5, 8]. The existing literature highlights the Global North's dominance in the production of climate change research [20–25]; IPCC author participation [5, 20, 26–30]; and intergovernmental involvement in the panel itself [31–34]. More recently, scholarship has also identified Saudi Arabia, China, and India as members of a small group of states influential in the approval of the assessments' key findings [5, 7–10].

Qualitative research on the IPCC approval process frequently relies on *Earth Negotiations Bulletin* (ENB) meeting summaries as a source of information on government intervention during the proceedings [35]. We build on this research by applying a flexible approach from natural language processing to quantify government engagement *by topic* as recorded in ENB text. ENB reports are not verbatim transcripts of IPCC sessions, but they are systematic summaries of the discussions during the plenary proceedings.While they may not cover each topic with the same level of detail or in a perfectly balanced way, contentious topics, such as carbon management and nature-based solutions (NbS), are well documented. For these "hot" topics, ENB reports reflect general patterns of government interventions and their standardized language and structured reporting format make them ideal inputs for quantitative, text-analytical analysis.

Our method uses word embeddings as a numerical representation of text which allows us to construct a text-based measure of each government's engagement levels. We obtain these from semantic differences between a (set of) topic-specific target word(s) and country mentions in written ENB meeting accounts. We illustrate this method for two example topics that received particular attention during AR6: carbon management—which we use as an umbrella term to refer to a set of abatement and removal technologies, including carbon capture and storage (CCS); carbon capture, utilization, and storage (CCUS); and carbon dioxide removal (CDR)—and nature-based solutions (NbS). We focus our analysis on these two topics because of their relevance during the IPCC approval plenaries and their explicit reference in the COP 28 Global Stocktake (GST) outcome decision, which calls on parties to "accelerating zero- and low-emission technologies, including (...) *abatement and removal technologies such as carbon capture and utilization and storage*" and to "accelerating the use of *ecosystem-based adaptation and nature-based solutions* [36, CMA.5, italics added].

Our method innovates on present scholarship in two important ways. Methodologically, it allows us to *numerically quantify* governments' engagement levels across different topics and rank them on a common scale. This works because even for countries with few mentions overall, but consistent mentions on the same topic, our method is able to detect these patterns in the text data. Substantively, the topic-specific variation in governments' engagement levels, which we document, helps nuance existing research that has identified several most active countries which appeared to dominate throughout the approval process [5, 8, 9]. We, however, show that the level of engagement depends on the topic under approval. While some governments, such as Norway, Germany, Saudi Arabia, India, and the U.S., engage consistently on both carbon management and NbS, other states primarily engage only on a single topic. For carbon management, these are the delegations from Japan, Luxembourg, and the Netherlands, while the most engaged governments on NbS are those of South Africa, Argentina, and Ecuador. According to our measures, engagement levels for NbS are 25-30% lower than for carbon management, which illustrates how our engagement scores can be used to assess differences in the relative intensities of government engagement across topics. We furthermore show that delegation size poorly explains variation in governments' engagement levels. Overall, we offer compelling evidence for meaningful topic-level variation in member governments' engagement levels during IPCC plenary sessions, which existing work has failed to recognize conceptually and measure with similar degrees of granularity.

These findings are consequential not only for advancing a more nuanced and robustly measured understanding of which governments engage on which topics in IPCC proceedings. They also matter as a foundation for future research into how engagement in the IPCC is tied to how particular topics can shape UNFCCC negotiated outcomes. In doing so, our work contributes to the growing research program that recognizes the important link between the IPCC and the UNFCCC [3, 5–7, 19, 37–42]. The robust and replicable, state-of-art method, which we introduce, is moreover not limited to IPCC research but can guide systematic analysis of topic-specific government engagement in any intergovernmental or negotiating process, where written meeting records are available.

Mapping governments' engagement levels by topic from text using word embeddings

Our method of mapping government engagement levels from written ENB summary reports builds on advances in natural language processing and the numerical representation of text as "word embeddings." Word embeddings represent the meaning of words (*semantics*) by assigning them a set of numbers and locating them in an embedding space, similar to how coordinates assign places to a longitude-latitude space on a map. Based on the linguistic premise that neighboring words share similar semantics [43, 44], our method estimates word embeddings for a (set of) target word(s) of interest from its surrounding words [45, 62]. We then use distances between target words to create our engagement measure in the following step-wise process (Fig 1).

1. First, we calculate the word embedding for each single occurrence of a target word (e.g., "CCS") from pre-trained embeddings of neighboring words (Step 1a)—taken from the "Global Vectors of Word Representation" project [46]. We then average target word embeddings across all instances in the ENB report to obtain a corpus-level embedding for the target word (Step 1b).
2. Our second step repeats the same procedure for a country name (e.g., "Japan").
3. We then calculate the engagement measure as the cosine similarity between the embeddings of the target word and the country name (Step 3).
4. We plot engagement measures for all countries in the ENB report on a $[-1, 1]$ interval (Step 4)

This procedure produces measures of government engagement with a particular topic from the *semantic distance* between target words (associated with that topic) and mentions of country names in ENB reports. High similarity scores indicate that target words and country names are either repeatedly mentioned next to each other (for example, "Japan" might often appear directly next to "CCS") or that countries are named together with words that share the target word's semantic space (for example, "Japan" might not appear directly next to "CCS" as such, but it might appear close to "direct air capture," which shares the semantic space with "CCS"). In either case, and validating our measure, semantic proximity between target words and country names can only be high when government delegations in IPCC meetings are recorded as engaging with a particular topic circumscribed by the same (set of) target word(s)—such as "CCS" as a target word for the broader topic of "carbon management."

We also obtain measures of uncertainty for our engagement scores. These allow us to distinguish whether a country's high/low engagement level with a given topic is simply because this country generally engages a lot/a little with IPCC negotiations. Countries with large delegations, for example, can engage more actively across many topics [47]. Our uncertainty measures account for such country-specific differences by computing placebo cosine similarities from 1,000 copies of the ENB summary texts for which we reshuffle the position of country mentions in the text. This permutation inference helps us rule out that our results are driven, for instance, by differences in countries' capacity to engage, or other systematic country-level variation. As a result, even if a country was only mentioned five times in the ENB reports across all four approval sessions in AR6, but all these mentions were in the context of carbon management, our method would assign a high engagement score for this country *on this very topic*.

Results

We show results from mapping governments' engagement levels in the approval of AR6 for carbon management and nature-based solutions as two topics that were identified as key policy approaches for national mitigation and adaptation plans in the negotiated Global Stocktake outcome of COP28 in Dubai, otherwise known as the UAE consensus. Before presenting topic-specific evidence, we contextualize our findings by reporting the distribution of overall government engagement—independent of topic—in IPCC approvals through a measure of country mentions in ENB summary reports. As previous studies highlight, engagement is heavily skewed [5, 8–10].

Based on ENB data from all four IPCC approval sessions in AR6, India (522 mentions), Saudi Arabia (379 mentions), the United States (309 mentions), Germany (251 mentions), and Norway (202 mentions) are the most mentioned countries, followed by France (161 mentions), China (133 mentions), Brazil (130 mentions), the United Kingdom (108 mentions), and Switzerland (107 mentions). Delegations from these ten countries account for almost 60% of all mentions, supporting previous

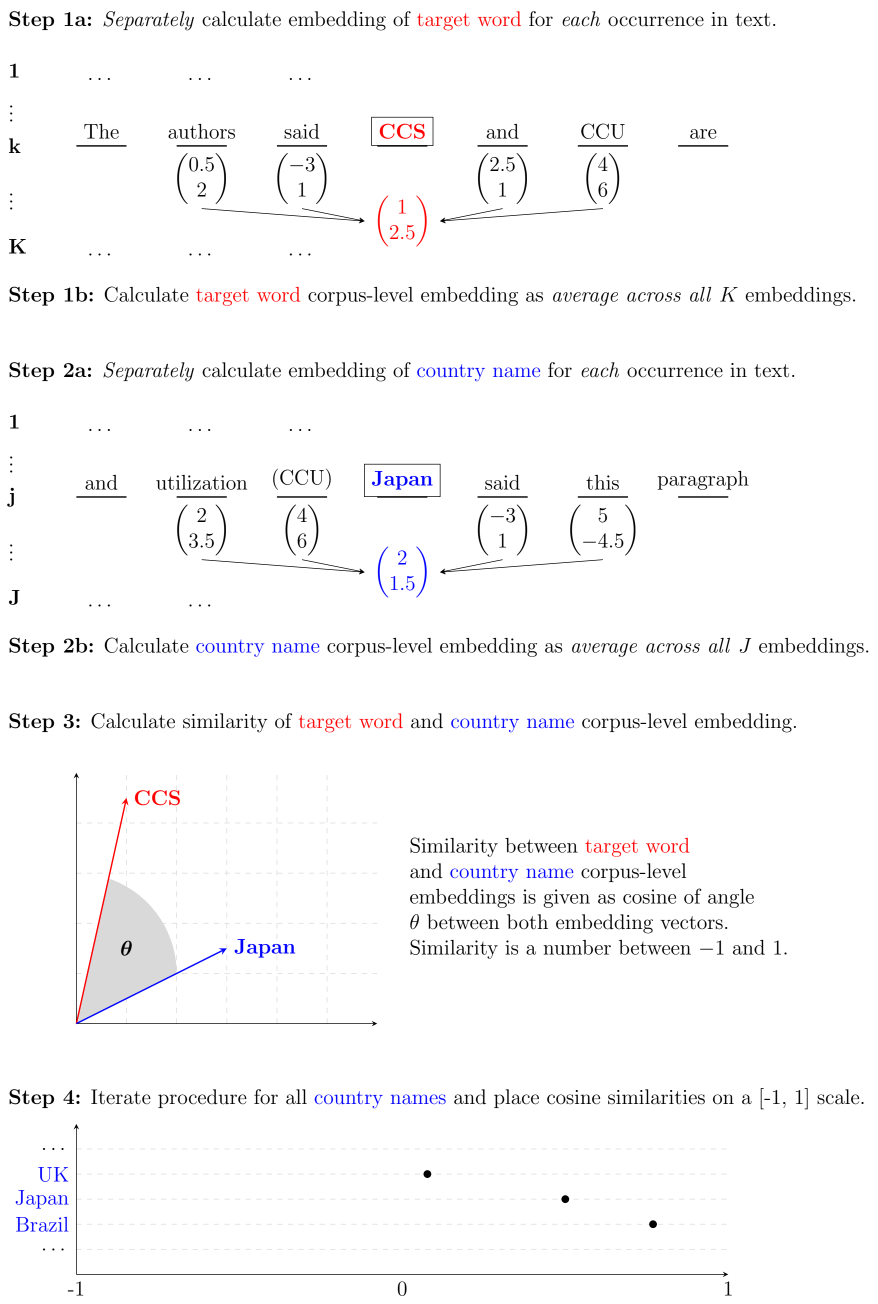

Fig. 1 Simplified illustration of how we map governments' engagement levels by topic from text with word embeddings. For *each* occurrence of the target word "CCS" (red), step 1a calculates the embedding of this target word as the simple average of pre-trained embeddings of the two words around it (black) and in two dimensions (numbers in brackets). Step 1b averages over all occurrences of the target word to calculate the corpus-level embedding of "CCS." Steps 2a and 2b repeat the same procedure for "Japan" (blue) as an example of a target country. Step 3 calculates the cosine of the angle between the embedding vectors for "CCS" (red) and "Japan" (blue) as our similarity measure. Step 4 iterates the procedure for all country names in the text and locates similarity scores for each of these countries on a scale between −1 and +1.
Note: Numbers, dimensions, and window sizes are for illustration. Our full analysis uses a transformation matrix to downweight common words, a 300-dimensional embeddings space, and 10-word window sizes each side of the target word (see Online Materials).

research that has identified a similarly small group of countries as most actively engaged in proceedings [5, 10]. While these overall patterns matter, our analysis points to largely overlooked variation in engagement levels by topic.

Results for the topic of carbon management

"Carbon management" as an umbrella term refers to a suite of abatement and removal technologies designed to capture, utilize, store, reduce, and remove CO_2 emissions. It has risen to prominence in international climate governance as a policy response for hard-to-abate sectors, to counterbalance residual emissions in net zero pledges, and to stay within the temperature targets of the Paris Agreement [37, 38]. However, carbon management comes with the risk of potential mitigation deterrence in the short term [48–50] and lock-in into high temperature futures in the medium term, especially if currently nascent technologies cannot be deployed at scale [51]. Concerns that these technologies encourage the continued extraction and use of fossil fuels have been voiced both in the context of climate negotiations under the UNFCCC, including during the Global Stocktake at COP28, and in discussions over scenario pathways in integrated assessment models in IPCC reports [37, 48, 51–53]. Carbon management, therefore, makes for a useful and highly salient topic for which to demonstrate our method.

Fig 2 presents the main results for the following set of target words that we use to characterize the topic of "carbon management:" carbon dioxide removal, carbon capture and storage, carbon capture and utilization, direct air capture and storage, bioenergy with carbon capture and storage, and their acronyms. Points indicate government engagement levels (measured as cosine similarity scores between target word embeddings and country name embeddings), which allow us to order member governments present in IPCC meetings from low engagement (top left) to high engagement (bottom right). For each country, we also plot a distribution in gray which shows the statistically expected range of engagement levels if governments were to engage with topics at random.

Based on estimated engagement scores, which we report in brackets, we find that the ten most actively engaged delegations in discussions on carbon management in IPCC approval meetings are Norway (0.895), France (0.891), German (0.878), Japan (0.874), Luxembourg (0.870), Saudi Arabia (0.867), Belgium (0.856), India (0.856), St. Lucia (0.855), and the Netherlands (0.855). All these countries engaged in much higher levels *on this topic* than what we would expect to see by chance alone (all statistically significant at $p < 0.05$). These results are interesting for multiple reasons. First, this ranking based on topic-level engagement scores differs from the one reported above, where we ordered countries simply based on the absolute number of their total mentions in the ENB reports. Second, other countries whose overall engagement levels fall into a medium range (lower than 100 overall mentions), such as Japan, Luxembourg, Belgium, St. Lucia, and the Netherlands, engage heavily on carbon management issues and rank among the top-10 based on similarity scores. Third, Switzerland shows muted engagement with carbon management issues despite active overall engagement levels (score of 0.805, ranked #21). While our method cannot tell us whether governments engage in supportive or obstructive ways on a given topic, it helps us quantify government engagement for a particular topic. This allows us to measure, assess, and compare variation in engagement intensity across different government delegations. Importantly, the same is true for the other end of the ranking as well, and particularly so for delegations marked with white dots (and red asterisks for statistical significance), which engage "too little" compared to what we would expect from our statistical model. Our method can therefore be helpful for identifying overly "silent" member governments and prompt further research into the motivations for this behavior.

For probing the face validity of our word embeddings approach, high engagement levels of St. Lucia (48 mentions overall, engagement score of 0.855 and rank #9 on carbon management) and St. Kitts & Nevis (75 mentions overall, engagement score of 0.851 and rank #13 on carbon management) are consistent with our own and other observer accounts that delegations of small island states have been particularly vocal on the topic of carbon management throughout AR6 by referring to associated technologies as "false solutions" [54] and querying their costs, feasibility, long-term operational safety, and their potential for mitigation deterrence. Similarly, the finding that Denmark ranks highly based on our engagement measure (engagement score of 0.848 and ranked #15th) is insightful because, although Denmark is only mentioned 45 times overall (1.1% of mentions) in the ENB summary reports, our method produces results that fit well with the Danish government's plan of advancing an industrial growth strategy centered around carbon dioxide abatement technologies [55].

Fig 3 plots a two-dimensional representation of the semantic space that underlies our results above (using Uniform Manifold Approximation and Projection (UMAP) for dimensionality reduction). The inset zooms in on the central part of the plot which is most relevant for our discussion. We find that countries placed on the bottom right in the above Fig 2 because of their high (and statistically significant) engagement levels, cluster rather centrally (except maybe for the Netherlands) around the

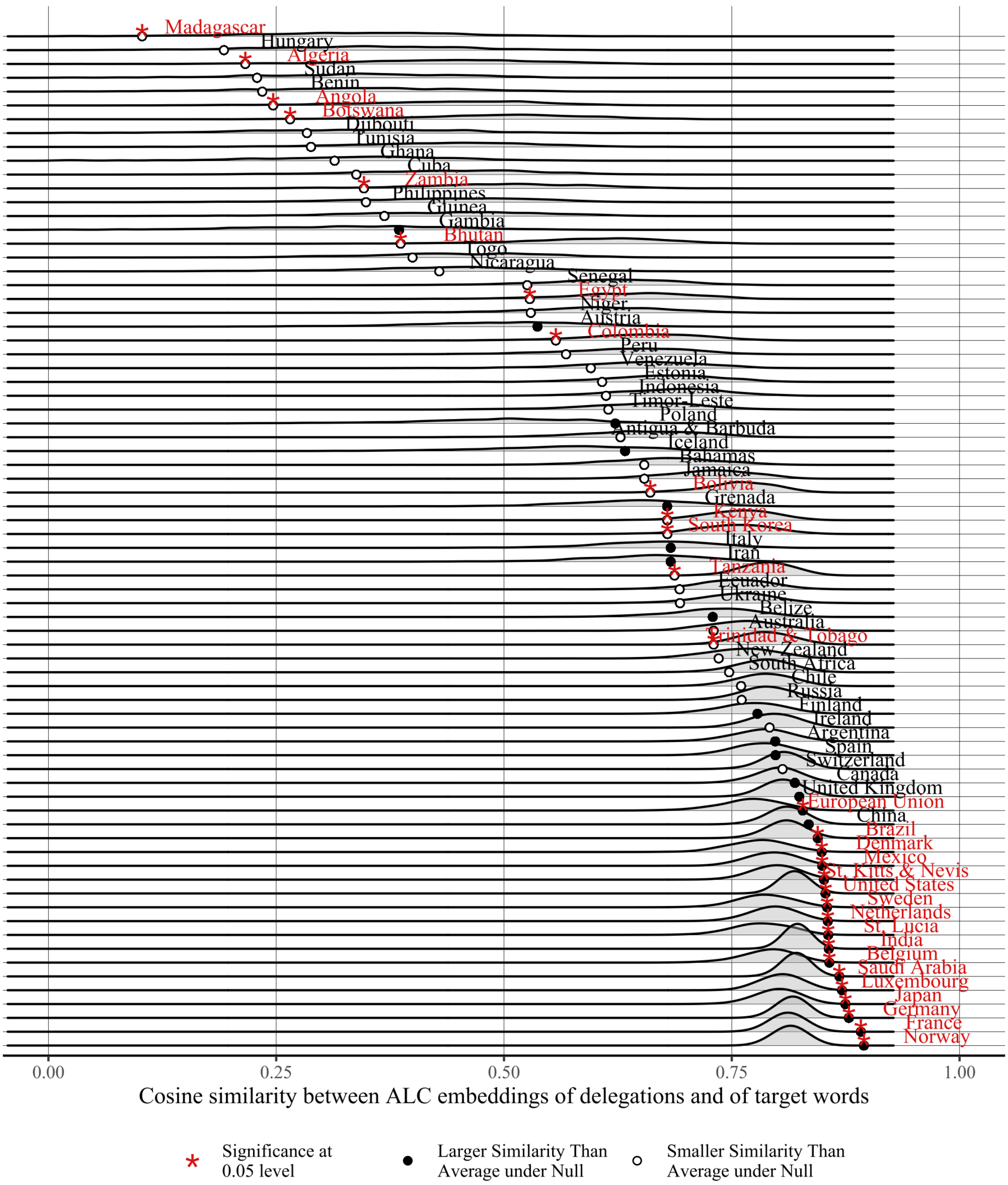

Fig. 2 Government engagement levels with the topic of "carbon management" across all IPCC approval sessions in AR6. Points show engagement levels, measured as cosine similarity scores between target words ("carbon dioxide removal," "carbon capture and storage," "carbon capture and utilization," "direct air carbon capture and storage," "bioenergy with carbon capture and storage," and their acronyms) and country names mentioned in ENB summary reports across all AR6 approval sessions. Densities report distributions of engagement levels under the assumption of random association between target words and mentions of country names. Country names and asterisks in red indicate engagement levels that are unlikely to have occurred by chance (statistical significance of $p < 0.05$, two-tailed test). Filled/hollow dots denote whether engagement levels are larger/smaller than the mean value of the distribution.

target words (shown in green) of "carbon dioxide removal," "CCS," and "carbon capture and utilization." Target words like "bioenergy with carbon capture and storage," "direct air carbon capture and storage," "BECCS," and "DACCS" are less centrally located and none of the countries shown in red, i.e., those with statistically significant engagement levels, can be found close to these words. Beyond country names and target words, words like "scenario," "phase-out," "overshoot," "near-term," or "limit" contextualize the semantic space from among a list of top-50 (nearest neighbor) vocabulary

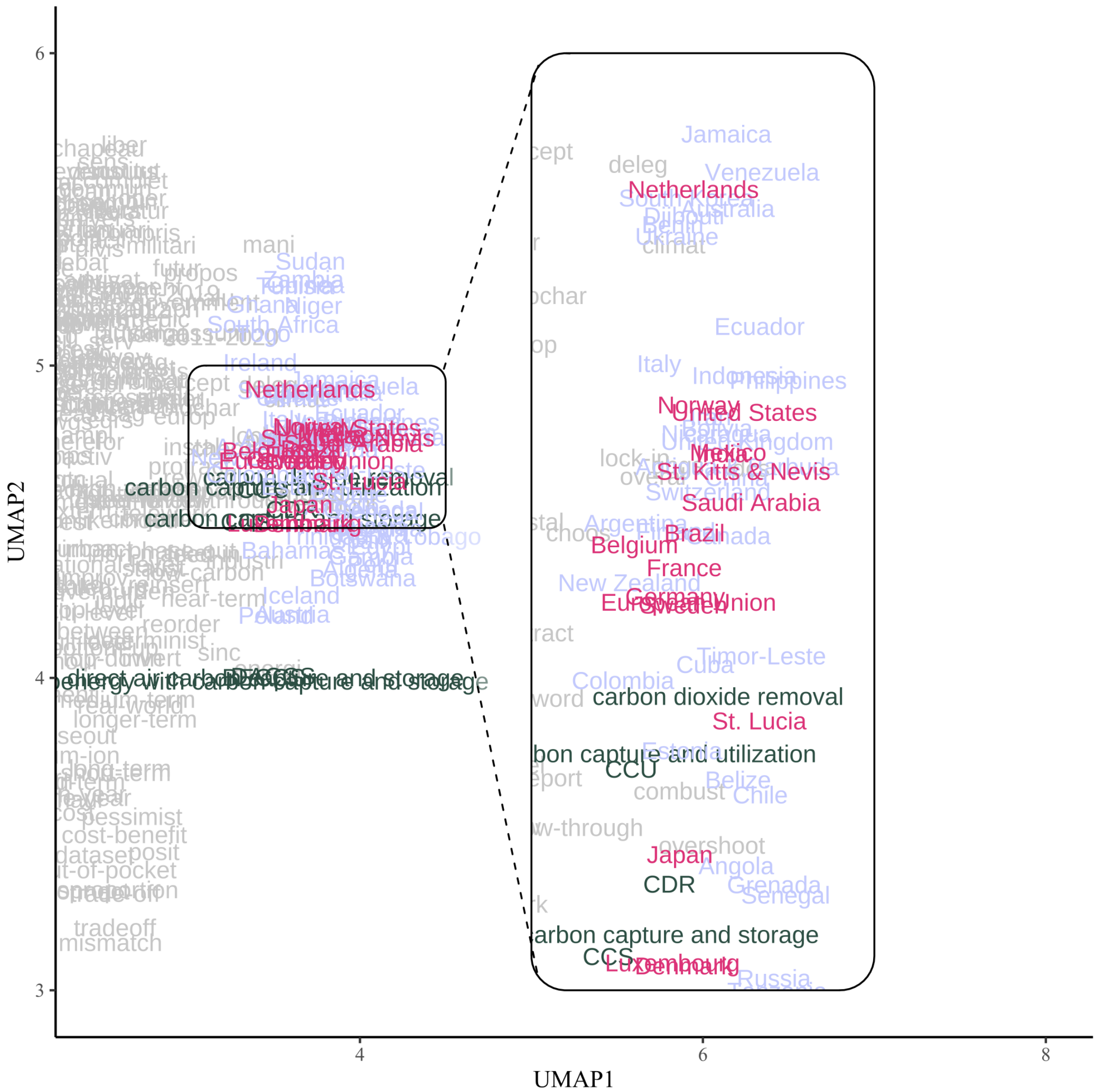

Fig. 3 Two-dimensional representation of the semantic space for the topic of "carbon management." This figure shows a truncated version of the semantic space of target words and country names for "carbon management" as the selected topic, based on the text corpus from ENB reports of all IPCC approval meetings in AR6. Words are located in 2-dimensional space after using UMAP dimensionality reduction with 15 neighbors around each word and a minimum cosine similarity distance of 0.1 to limit excessive clustering. Colors denote different types of words: target words are shown in dark green; country names in red indicate countries with statistically significant engagement levels with the topic, while country names in lilac are all other countries mentioned in the ENB report; words in gray are pre-trained embeddings.
Note: For better visualization, we truncated the semantic space to the part most relevant for our description of "carbon management" as the topic of interest. Appendix B shows the full semantic space. The inset zooms in on the central part of the plot for readability.

(S1 Appendix shows a full list). This supplementary analysis of the semantic space offers descriptive evidence from our text-based method that, among the set of target words to characterize the wider topic of carbon management, aspects around CDR, CCS, and CCU are linked to statistically significant engagement levels.

Results for the topic of nature-based solutions

We also apply our method to the topic of nature-based solutions (NbS), which are defined by the IPCC AR6 Working Group II Glossary as "actions to protect, sustainably manage and restore natural or modified ecosystems that address societal challenges effectively and adaptively, simultaneously providing human well-being and biodiversity benefits" [56]. The term has gained traction because of the co-benefits it offers to both climate change adaptation and mitigation and in addressing biodiversity loss. While there is growing scientific and political interest in NbS, and some governments have incorporated NbS approaches in climate action plans, concerns have been raised over competition with other land uses, reliance on monocrop forestry and delay of the energy transition from fossil fuels [57, 58]. It has also been argued that deliberately framing such solutions as "natural" may leave a false impression that NbS are necessarily more feasible, cost-effective, desirable or democratic than other climate solutions [59, 60]. As with abatement and removal technologies, NbS appears in the final outcome of the GST.

However, it differs from the topic of carbon management in interesting and important ways that allow us to demonstrate both the strength of ENB summaries as source material and our mapping-by-topic method. The abatement and removal technologies captured under the "carbon management" topic analyzed above, received extensive coverage in the Summary for Policymakers of Working Group III and appeared in the final Synthesis Report of the AR6. NbS on the other hand, receives a total of 45 mentions in the ENB reports of the approval of the AR6 and only appears as a footnote in any of the SPM documents. This allows us to identify the interest and engagement in a topic that has ultimately largely been removed through the approval process.

We plot the topic-specific results for NbS from nature-based solutions, ecosystem-based adaptation, and their acronyms as target words in Fig 4. The ten most active delegations according to estimated engagement scores, shown in brackets, are South Africa (0.792), Argentina (0.744), France (0.709), the EU (0.699), Belgium (0.682), Brazil (0.681), Germany (0.679), Ecuador (0.676), the U.S. (0.671), and Norway (0.665). All these engagement scores are statistically significant at 5% levels, indicating greater government intervention on this topic than when interventions were to happen at random. Aside from mostly European countries and the United States with high engagement activity across both topics, our method places South Africa and Argentina on top of the NbS ranking, with Brazil and Ecuador as two other major South American forest nations featuring in sixth and eighth place, respectively. Apart from Brazil (130 mentions, 3.3% of total), all these countries display low overall engagement based on ENB mentions (South Africa, 49 mentions, 1.2% of total; Argentina, 47 mentions, 1.1% of total; Ecuador, 26 mentions, 0.6% of total), but are closely associated with NbS as a distinct topic, based on our method. These findings gain further validity in the context of IPCC plenary discussions during which South American countries and forest nations expressed concerns that their preferred (and through the Convention on Biological Diversity internationally recognized) approach of ecosystem-based adaptation is being displaced with a focus on NbS instead—a concept that was primarily elevated by European countries and the EU.

Comparison across topics

So far, our analysis focused on studying government engagement for a specific topic separately, i.e., either carbon management or nature-based solutions. Engagement scores, however, operate on the same scale and are, therefore, comparable across topics. This allows calculating simple descriptive statistics, such as means or standard deviations to characterize differences across topics. In our case, for example, average engagement for carbon management (mean=0.630) is higher than for NbS (mean=0.504) and the range between the 25% and 75% percentile of engagement scores for carbon management ([0.562,0.829]) is considerably shifted upwards compared to the topic of nature-based solutions ([0.389,0.630]). These differences matter because they allow a quantified assessment of relative engagement intensities. For carbon management, the top ten countries' engagement scores are similar, while they are much more spread out for NbS. The difference between a score of 0.895 for Norway (ranked #1) and 0.855 for the Netherlands (ranked #10) is small, while the difference for NbS scores between South Africa (score of 0.791, ranked #1) and Norway (score of 0.665, ranked #10) is almost three times as large. This points to greater engagement activity and to the fact that, by this metric, carbon management was a more contested topic in comparison to NbS.

Fig 5 shows the distribution of engagement scores as histograms on the margins and visualizes these relationships. The scatter plot maps engagement scores for carbon management and NbS along the horizontal and vertical axis, respectively, together with a 45-degree line, which indicates equal engagement on both topics. Most countries, with the exception of South Africa and large forest nations like the Philippines or Nicaragua, are located below this line. These countries therefore engage relatively more on carbon management than on NbS. The plot also captures the clustering of several countries heavily engaged in carbon management in the north-east quadrant of the figure. This

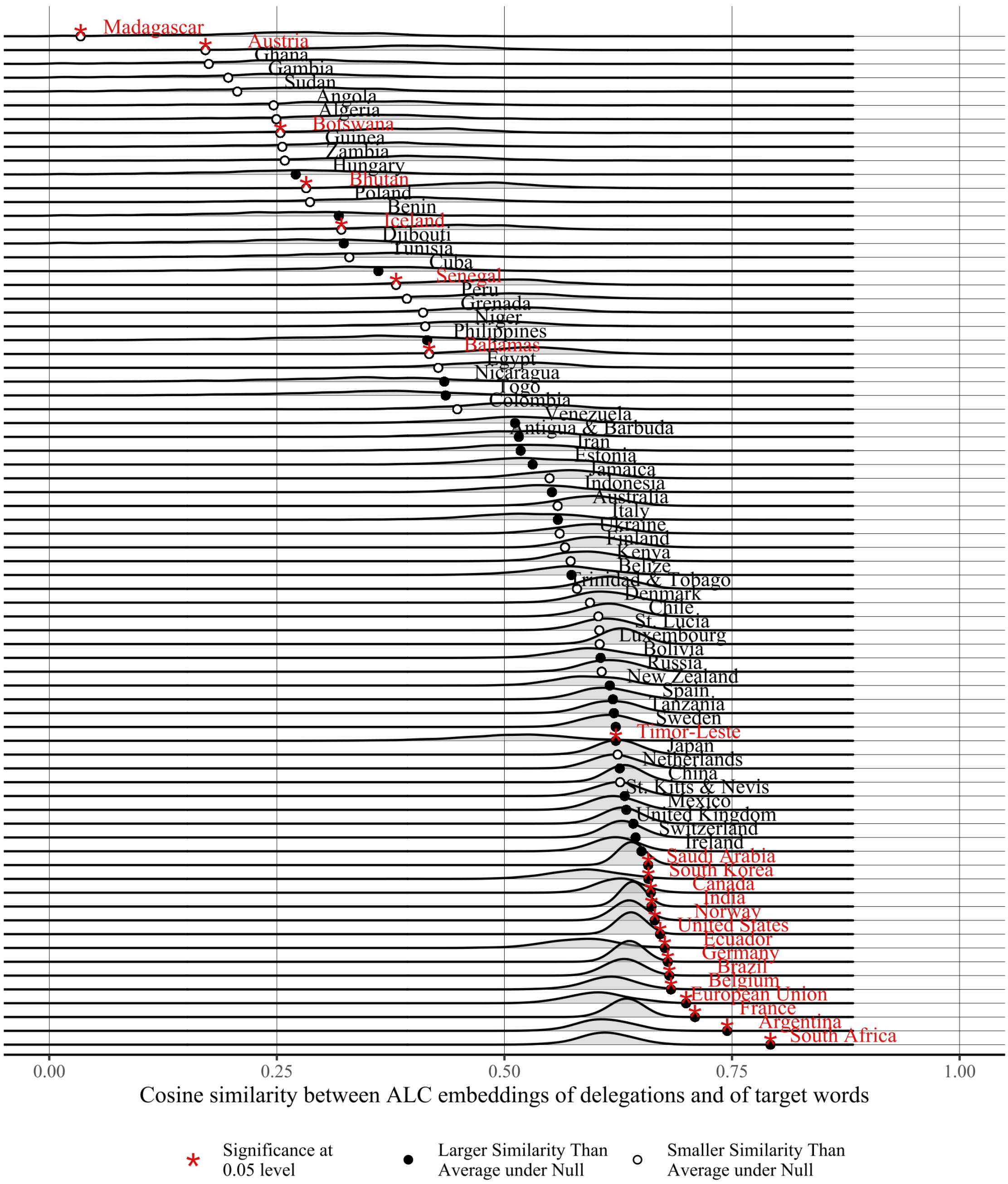

Fig. 4 Government engagement levels with the topic of "nature-based solutions (NbS)" across all IPCC approval sessions in AR6. Points show engagement levels, measured as cosine similarity scores between target words ("nature-based solutions," "ecosystem-based adaptation," and their acronyms) and country names mentioned in ENB summary reports across all AR6 approval sessions. Densities report distributions of engagement levels under the assumption of random association between target words and mentions of country names. Country names and asterisks in red indicate engagement levels that are unlikely to have occurred by chance (statistical significance of $p < 0.05$, two-tailed test). Filled/hollow dots denote whether engagement levels are larger/smaller than the mean value of the distribution.

illustration helps assessing the relative intensity of government engagement both across different governments on the same topic and for the same government across different topics.

Engagement levels and delegation size

Lastly, we demonstrate that the variation in topic-specific engagement that we identify is not simply explained by differences in delegation size. It seems intuitive that countries with larger delegations have a greater capacity to engage in IPCC approval sessions, especially across multiple topics, but

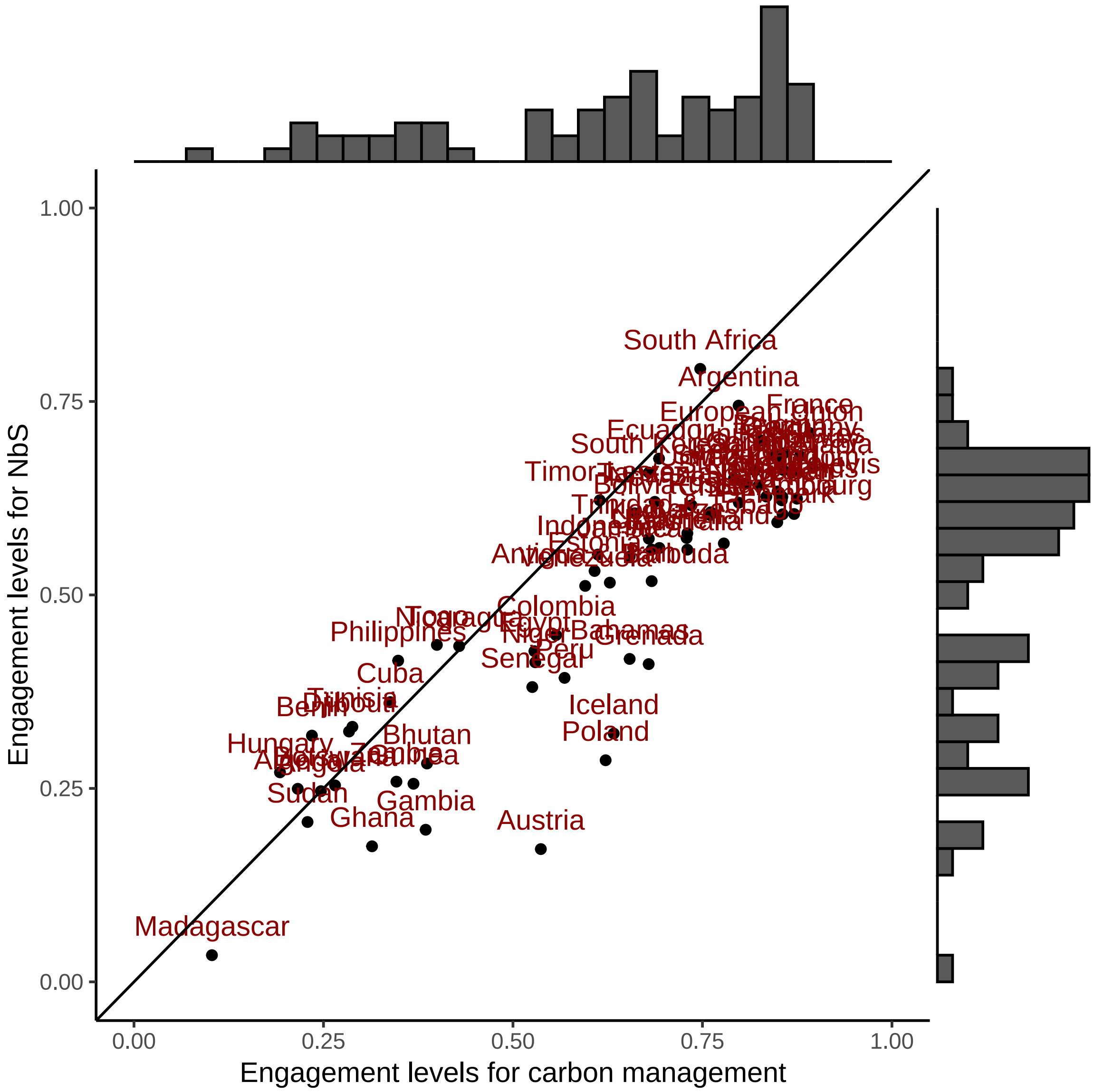

Fig. 5 Scatter plot of government engagement levels for carbon management and NbS. Points show government delegations' engagement levels for carbon management and NbS on the horizontal and vertical axis, respectively. Histograms on the margins show full distribution of engagement scores by topic. 45-degree line denotes equal engagement on both topics.

we find there is little evidence for this logic. For carbon management (blue) and NbS (orange), Fig 6 plots differences between actual (measured as cosine similarity) and predicted engagement levels, where predictions are modeled as a linear function of governments' logged average delegation size across WGI-III for AR6.

The plot visualizes three important patterns. First, for all countries and both topics, the differences between actual and predicted engagement levels, as indicated by the lollipop bars, are stark. A country's delegation size is therefore a poor proxy for making inferences about how actively governments contribute in IPCC approvals. Second, even for large delegations, such as the United States, the United Kingdom, China, or India, there is substantial *within* country variation. This means that even the most influential countries are not equally controlling of all topics. Third, and underpinning our central argument that governments' engage selectively on different topics, we identify three distinct groups of countries. Basing expectations of engagement levels on the size of a government's delegation size, some states, such as Algeria, Botswana, and the Philippines engage less than expected for both carbon management and NbS, but topic-level variation remains. Other countries like Jamaica, St. Kitts & Nevis, and St. Lucia engage more than expected on both topics, and generally more on carbon management than on NbS. And finally, there are countries that engage more than expected on

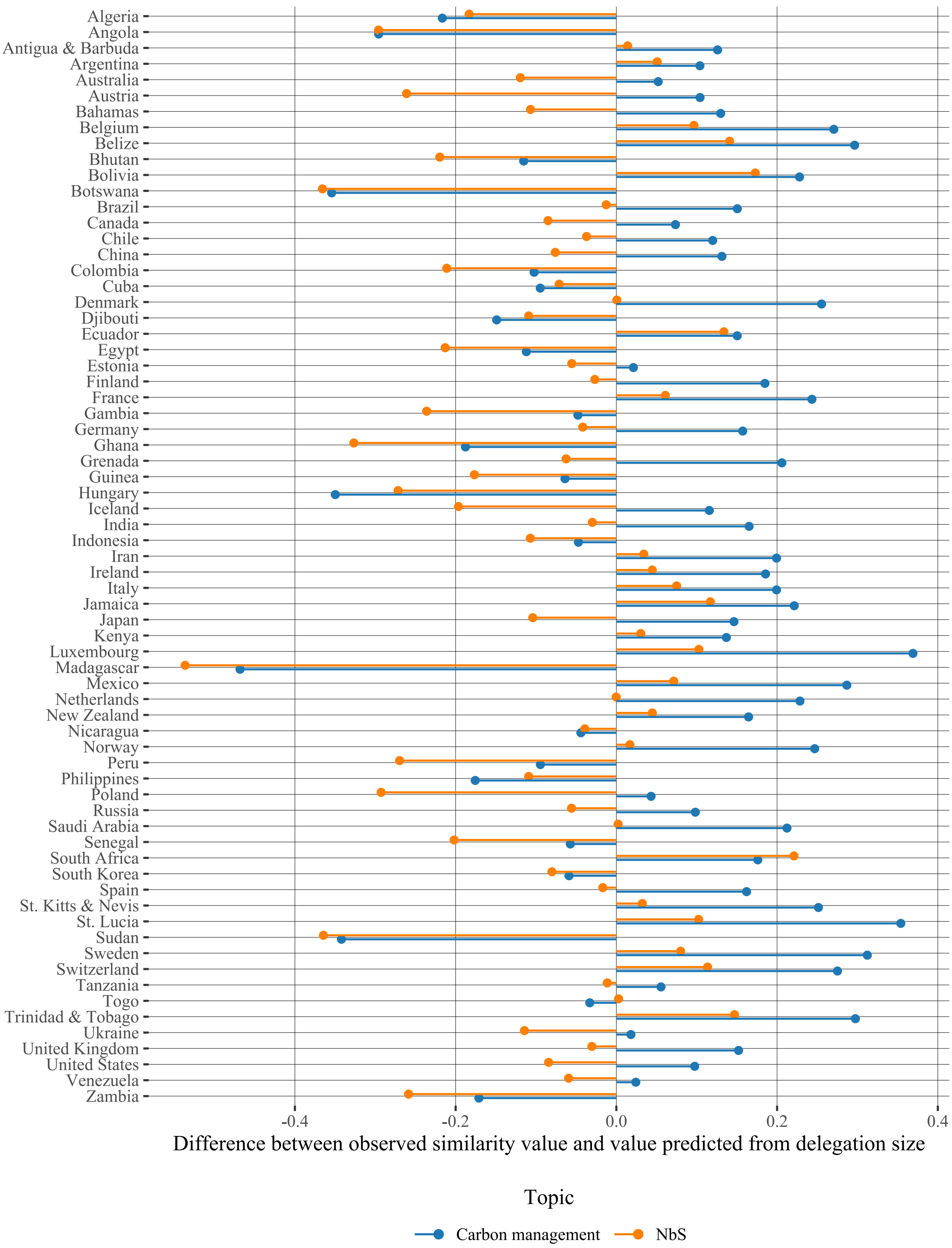

Fig. 6 Lollipop plot of differences in actual and predicted engagement levels by topic. The plot shows differences between actual and predicted engagement levels for the topics of carbon management (blue) and NbS (orange). Predicted values are from a linear regression that models engagement as a function of delegation size. The large deviations from the zero line indicates that delegation size is a poor predictor of member governments' engagement levels.

one topic and less than expected on the other one. This group covers a varied set of states, including the large delegations of Germany and Japan as well as the smaller ones of Venezuela or Grenada.

Conclusion

Existing research has demonstrated that government involvement in the IPCC is skewed towards a small number of influential countries [5, 8–10]. This finding derives mostly from ethnographic methods, including participant observation, interview data, and qualitative analysis of IPCC documents and ENB reports. Complementing these methodological approaches, we show how word embeddings from natural language processing can be used for mapping and quantifying governments' engagement levels by topic. Building on this text-as-data approach, we derive a measure that can help researchers analyze *to what extent* and *on what topic* governments engage, and how engagement levels *vary across these topics* under approval in the IPCC. The proposed method is versatile in two important ways as it can be applied to other central issues in climate policy debates and other intergovernmental, negotiating processes where written meeting records are available.

Carbon management, in the form of abatement and removal technologies, and nature-based solutions, which also include land-based approaches to carbon dioxide removal, were important topics of discussion in the approval of AR6. They also mattered during the technical and political phases of the GST in the UNFCCC and as policy options in national mitigation and adaptation plans for delivery on the Paris Agreement goals. Our results demonstrate that while there is a number of countries that engage in discussions on both topics, other governments engage much more *selectively*. We find that the delegations of Norway, France, Germany, Saudi Arabia, and India are heavily engaged on carbon management and, albeit with lower intensity, on NbS as well. In contrast, Japan, Luxembourg, Belgium, St. Lucia, and the Netherlands engaged almost exclusively on carbon management alone. South Africa, Argentina, Brazil, and Ecuador, on the other hand, show the opposite pattern and engage primarily on the topic of NbS. Based on the engagement scores that our method produces, carbon management, with higher and more clustered scores (0.895–0.855), was the more contested topic compared to NbS (0.792–0.665) in AR6. These results point to meaningful topic-specific variation in governments' engagement levels, which, as we show, is poorly explained by delegation size. Future research should therefore focus on advancing our knowledge of what drives topic-level engagement in the IPCC and its relationship to the negotiation of climate mitigation in the UNFCCC.

Our findings also matter in the context of AR7. Robust evidence that countries engaged differently on two topics in the approval of key SPM findings should draw attention among member governments to the importance of investing in IPCC participation alongside the UNFCCC. Active engagement in the production and approval of IPCC products ensures the relevance and uptake of the final assessments to diverse national contexts. Engaging in the approval process provides all governments with a critically important opportunity to increase their knowledge base on central topics as well as the scientific and political debates that surround them. It also invites participation in the co-crafting of these topics with IPCC authors before they are negotiated in the UNFCCC.

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